



CHRONIC CHANGES IN THE RABBIT TIBIOFEMORAL JOINT FOLLOWING A SINGLE BLUNT IMPACT

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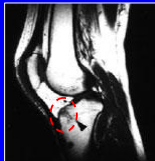
INTRODUCTION

Constrained



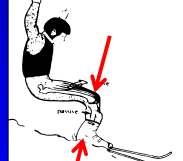
Bone fracture @ 11xBW¹

- Bone bruises in 80% of patients with ACL tears.³



Types Of Tibiofemoral Injuries

Unconstrained



ACL rupture @ 6-7xBW²

- Cartilage fissuring in areas overlying bone bruises.⁴



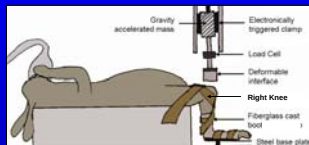
HYPOTHESIS

- A single impact to the tibiofemoral joint, with less force than is required to generate gross injuries such as bone fracture or ACL rupture, will result in development of degenerative joint disease.¹

METHODS

- Flemish Giant rabbits- 10 impacted + 10 control animals.

- Right leg impacted.



- Exercised 10 min/day 5 day/week.⁵

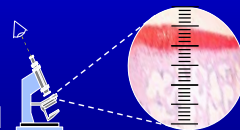


- Sacrificed after 12 months.

- Mechanical properties measured at 4 locations.

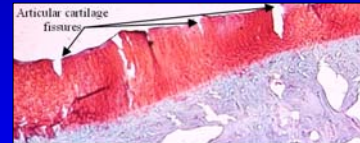


- Gross and histological photographs were analyzed.

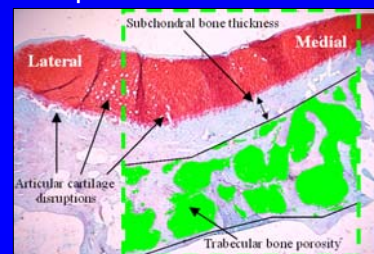


RESULTS

- No gross injuries from impacts.

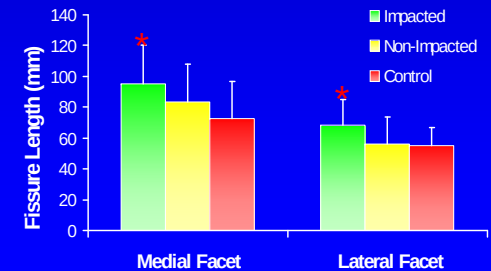


- Significantly more tissue disruptions for impacted and nonimpacted vs control limbs.



- 15% increase in subchondral bone thickness for impacted vs nonimpacted and control limbs.
- 5% increase in trabecular bone porosity for impacted vs control limbs.

- Significantly more fissures for impacted vs nonimpacted and control limbs.*



- 12% reduction in the cartilage elastic modulus at sites 1 & 3 for impacted vs control limbs.

CONCLUSIONS

- The fissuring and slight softening of impacted vs control cartilage were similar to other impact models⁶ and reminiscent of early clinical OA.⁷
- Changes in the subchondral and trabecular bone underlying the impacted joint cartilage were consistent with degenerative joint disease induced by ACL transection.⁸
- A slight degenerative effect was noted in the opposite, non-impacted knees in this model and may be related to altered loading patterns or a systematic effect of the trauma.⁹
- Finally, the thickening of subchondral bone may help explain trabecular bone changes by a stress shielding mechanism in this model.¹⁰

REFERENCES

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